



Company Portal Deep Dive

From Opening to Installing



“

Speaker for today



Rudy Ooms

”

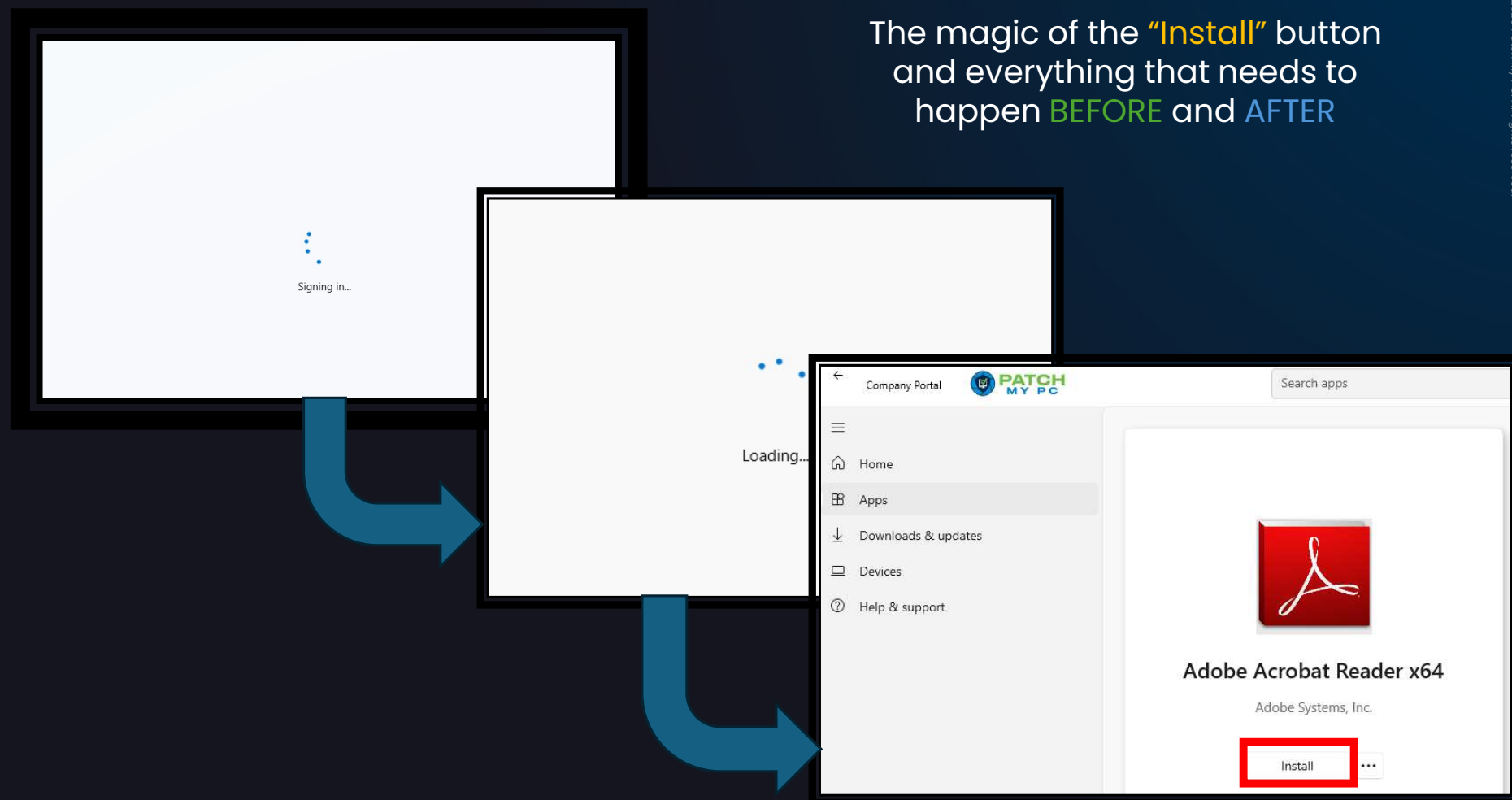


The Company Portal



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The magic of the **"Install"** button and everything that needs to happen **BEFORE** and **AFTER**





Agenda

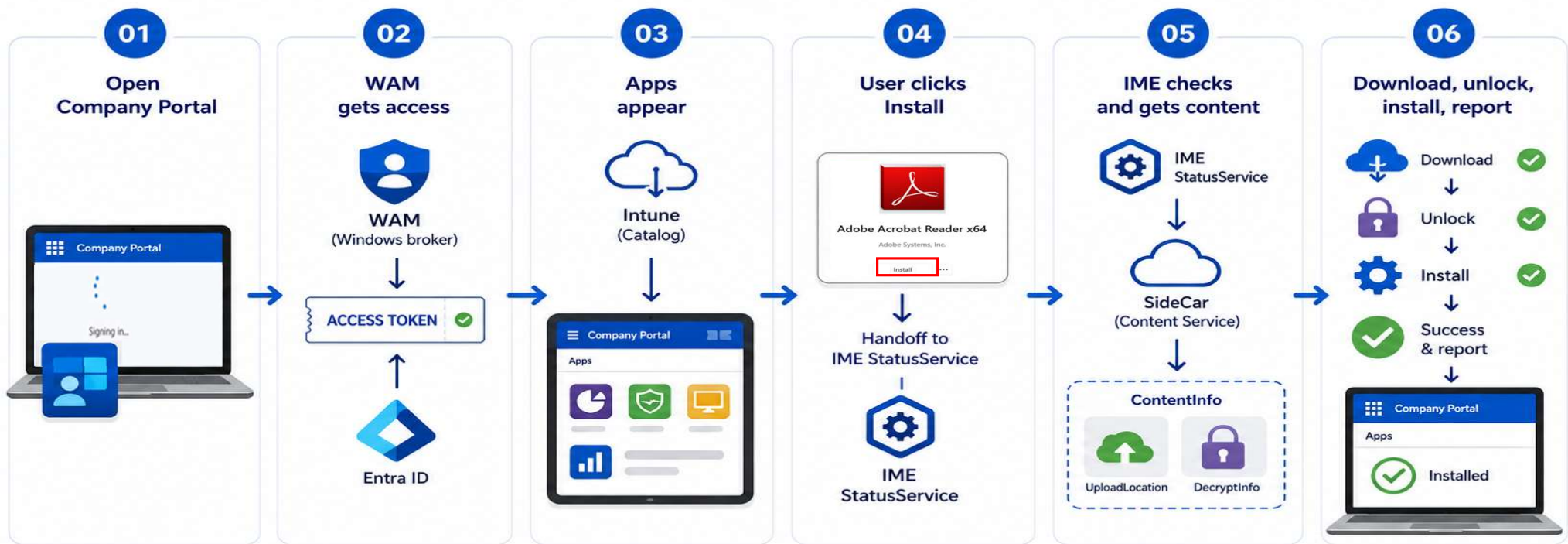
- The Full Flow
- The Company Portal Signing In
- WAM not Wham!
- Apps Seem to Appear
- The Install Handoff
- The Download/Decryption

The Full flow....

(please pay attention)



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One click starts a chain: access, catalog, handoff, content, install, and reporting.

Open Company Portal

SSO kicks in, you don't need to enter your credentials (WAM Kicks in!)

Apps appear

All the apps that are available are loaded (Metadata Catalog)

Click Install

With a simple click the user can trigger the install. The install is handed over to the IME

App Installed

IME tries to detect the app and reports the result.



The “Text” Version

Open Company Portal

SSO kicks in, you don't need to enter your credentials

Apps appear

All the apps that are available are loaded

Click Install

With a simple click the user can trigger the install

App Installed

IME reports the result.

Hidden underneath: (what I will explain in detail)

- WAM gets a token for the App catalog service and SideCar Service
- The token is sent over to get the App Metadata (Assigned Available Apps Information)
- The InstallAppRequest is sent over to the StatelessIWService (Information Worker Service)
- StatusService kicks off App Checkin and monitors the status
- SideCar returns package instructions
- DecryptInfo and the device key unlock the package

Before we even click on Install... there is a lot happening in the background... let's take it apart one by one

0.1 Company Portal Signing in

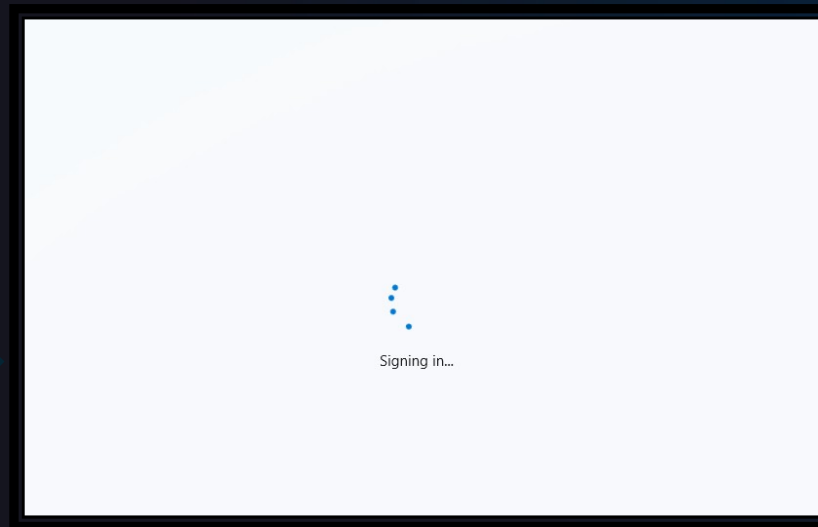
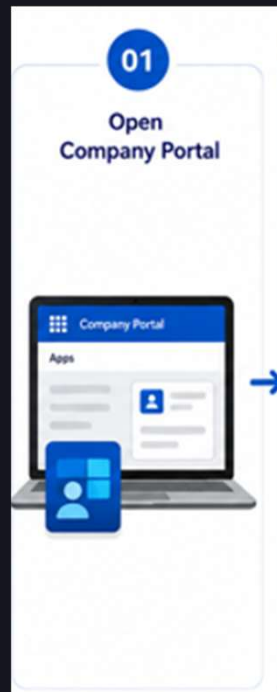


Company Portal Signing In

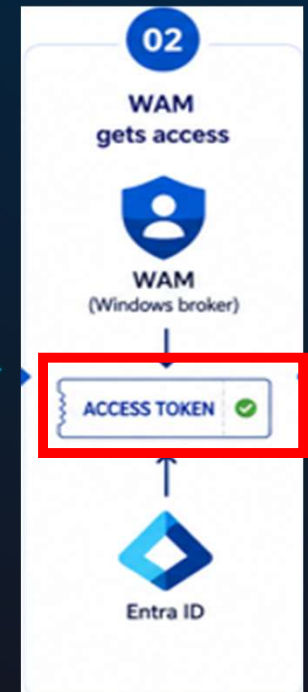
The user only notice the Signing In. The Company Portal is already asking for access.

1. Company Portal opens

When opening the CP, we need to perform authentication



To perform Authentication, we need a valid token to communicate with the services



02. WAM not Wham!





WAM: The Authentication Broker

The Company Portal Uses WAM (Web Account Manager) to get a “Badge” for the requested Services during sign-in

2. WAM gets involved

Windows helps request a token for the services.

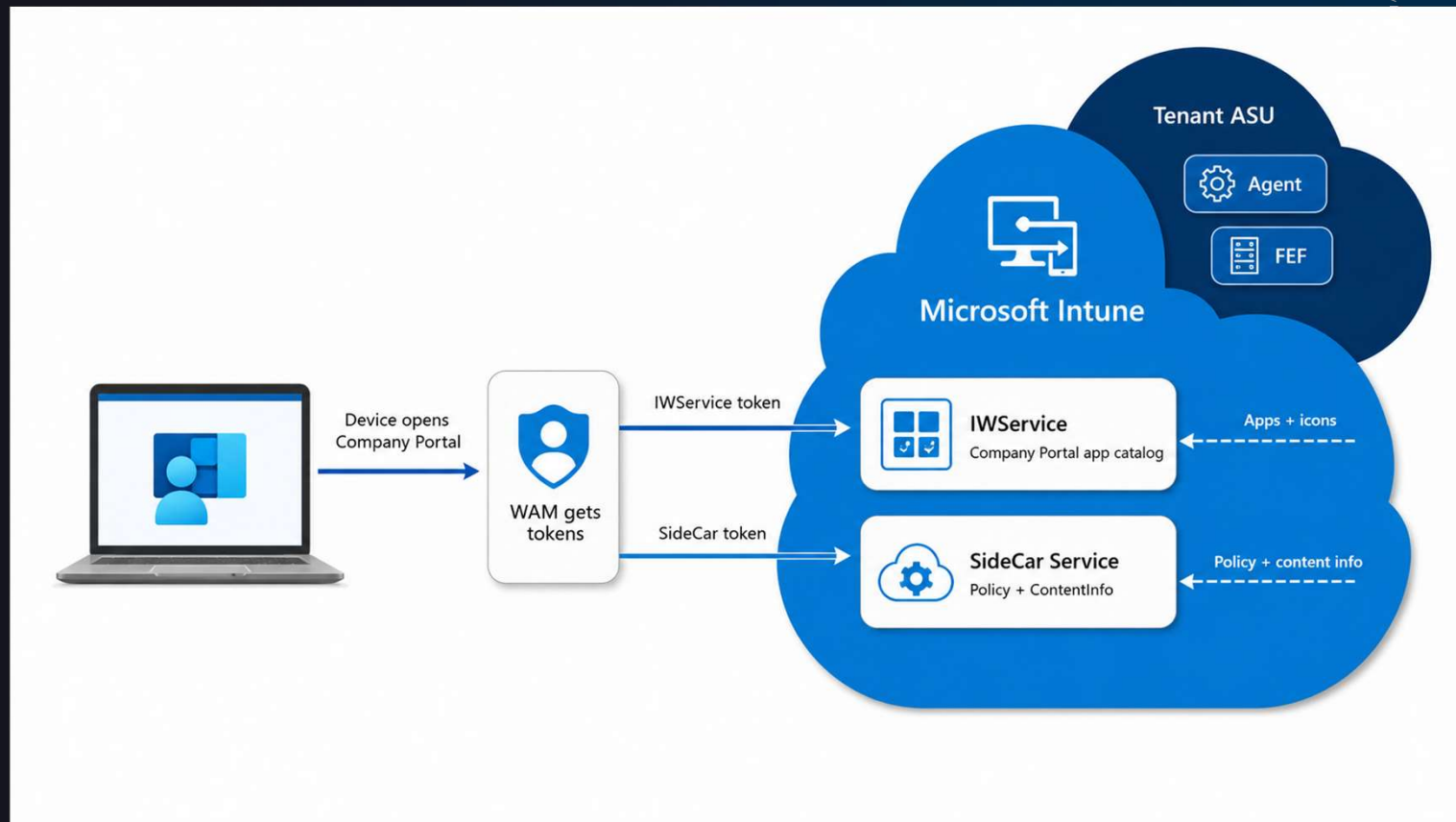
- WAM checks who the user is.
- WAM uses the user’s work account. Including PRT backed SSO
- WAM requests an access token for the specific requested service.
- WAM gives the token back to the app.
- The “App” can now connect without asking the user to sign in again.



We need 2 Tokens

The IWSERVICE and SideCar Service require authentication

The Company portal requires 2 tokens to get the “job” done!!!



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Different Rooms

Different services need different tokens



The 'IWService' token is required to retrieve all the Available Apps and its metadata.

The IME uses the SideCar token to request the Application data from the SideCar service.

IWService: Company Portal → loads the available apps and icons
SideCar: IME → gets policy and content info



The SideCar Token

The IME also needs a separate bearer token....

How does that Work?

The Company Portal asks WAM for a token scoped to the SideCar Service

We need this token to Authenticate against the SideCar Service! 26a4ae64...

Requesting the IME SideCar token through WAM

```
-----
Client application: Microsoft Intune Windows Agent
Requested resource: Intune SideCar / Check-in service
[VERB][WAM] Loaded C:\Windows\Microsoft.NET\Framework64\v4.0.30319\System.Runtime.WindowsRuntime.dll
[VERB][WAM] WinRT types loaded.
[INFO][WAM] Finding the Microsoft Entra WAM provider.
[VERB][WAM] Trying provider authority https://login.microsoftonline.com/common
[INFO][WAM] Provider found: Work or school account
[VERB][WAM] Provider authority: https://login.microsoftonline.com/common
[INFO][Identity] Client ID: fc0f3af4-6835-4174-b806-f7db311fd2f3
[INFO][Identity] Resource: 26a4ae64-5862-427f-a9b0-044e62572a4f
[INFO][Identity] Correlation ID: 6f21f2d4-290d-49b2-bad3-9f0b9092e535
[VERB][WAM] Resource set through WebTokenRequest.AppProperties using IDictionary<string,string>.
[INFO][WAM] Calling GetTokenSilentlyAsync.
[INFO][Identity] WAM returned the token silently.
[INFO][Identity] WAM account: rudy@patchmypc.com
```

```
Result      : WAM returned an IME SideCar token silently
User        : rudy@patchmypc.com
Audience    : 26a4ae64-5862-427f-a9b0-044e62572a4f
ExpectedAudience : 26a4ae64-5862-427f-a9b0-044e62572a4f
AudienceMatches : True
TenantId     : b79783df-131e-4d00-8303-6f66c0fa4bfb
CallingApplication : fc0f3af4-6835-4174-b806-f7db311fd2f3
ExpectedClientId : fc0f3af4-6835-4174-b806-f7db311fd2f3
IssuedAt      : 6/10/2026 11:38:50 AM
Expires       : 6/10/2026 12:45:53 PM
TokenLength   : 2206
```

WARNING: The complete token is sensitive.

eyJ0e
tYtli
4cCI6
aTjIz
yVkdY
llWRl
zOGVi

IsImtpZCI6IndoMDZzRwt6TE
zFlLTRKMDAtODMwMy02ZjY2Y
2JYdUlreFdxMU9rYng3QVU0N
UWoTnFFWThKUnJaY0RSdm1Uc
i1mN2RiMzExZmQyZjMiLCJhc
XAiOiJ1c2VyIiwiaXBhZGRyI
CJwd2RfdXJsIjoiaHR0cHM6L

03. Apps Appear

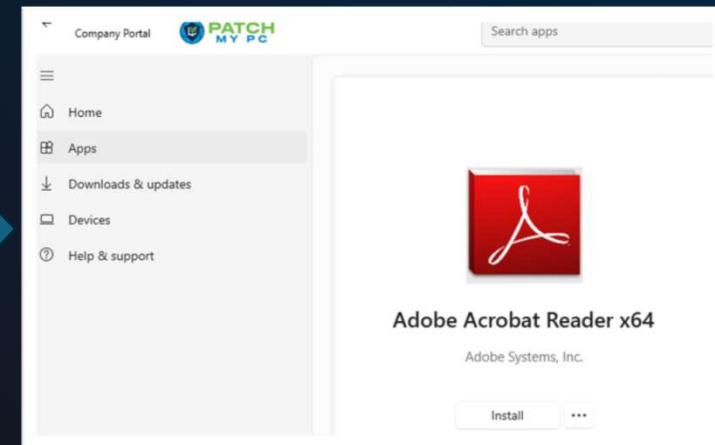


What comes after the WAM

3. Catalog can be called

Company Portal can now ask which apps it may show.

Loading...



The Token is then sent over to the IWService



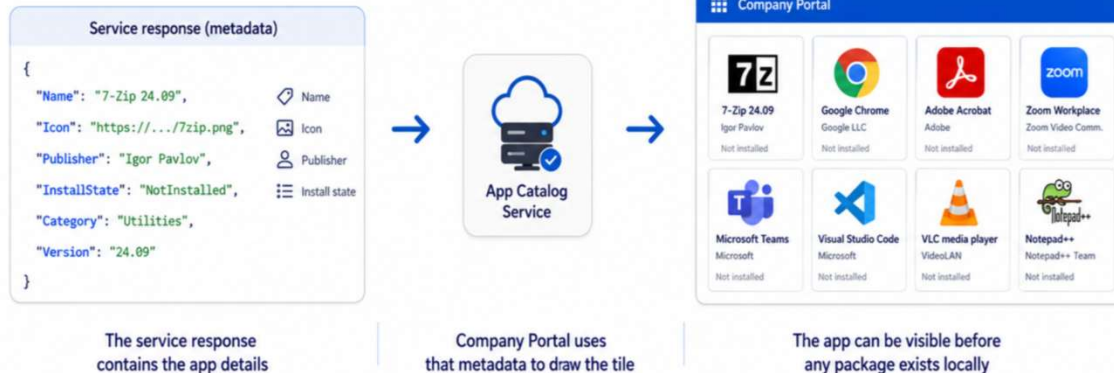


The MetaData = App Information

- 01 Open Company Portal
- 02 WAM gets access
- 03 Apps appear
- 04 User clicks Install
- 05 IME checks and gets content
- 06 Download, unlock, install, report

The answer becomes the app tiles

Metadata becomes a user-facing app tile



A visible tile means the catalog flow worked.
It does not mean the installer content is already on the device.

GET /TrafficGateway/.../IWSERVICE/Applications

Authorization: Bearer <catalog token>
(the token that WAM just got us)

Purpose:
return app metadata for this user/device

The returned IWSERVICE metadata (JSON) contains the information to show the app information and the fancy icons



The IWService....

Which was exactly what was broken last week

Error loading apps

Could not load apps due to a network issue. Check your connection and try again.

Share details

Close

• Jun 19, 2026, 6:30 PM GMT+2

We've identified that a portion of service infrastructure that facilitates loading apps in the Intune Company Portal for Windows and the web wasn't receiving request traffic as expected and caused impact. While we're continuing to investigate the reason it stopped receiving request traffic, we've confirmed that redirecting the request traffic to a functioning portion of infrastructure remediated the impact.

This is the final update for the event.

If there is an issue with the IWService, that required metadata will NOT be returned!

With it the Company Portal will fail loading the Apps!

GET request to

fef.amsub0102.manage.microsoft.com/TrafficGateway/TrafficRoutingService/IWService/StatelessIWService/Applications. Status:500 (Internal Server Error)

04. The Install Handoff



The Install Handoff

What happens if we click "Install"

4. The Company Portal turns the 'click' into install intent.

It sends the install request to IWSERVICE
with the IWSERVICE token
This request tells Intune, that a certain device
NEEDS to have that app installed... NOW!!!

```
2 references
private async Task SendIWSERVICEInstallRequestAsync(DownloadContext ctx, AppRow app)
{
    var token = await GetWamTokenAsync(CompanyPortalClientId, IWSERVICEResource, "AcquireIWSERVICETokenForInstallRequest");
    var baseUrl = GetIWSERVICEBaseUrlFromSideCarEndpoint(ctx.Endpoint);
    var iwSERVICEDeviceKey = await ResolveIWSERVICEDeviceKeyAsync(baseUrl, token, ctx.DeviceId);

    Log($"Sending IWSERVICE install request for available app [{app.Name}] [{app.Id}].");
    Log($"Local MDM certificate device id: {ctx.DeviceId}");
    Log($"IWSERVICE ApplicationState device key: {iwSERVICEDeviceKey}");

    var attempts = new List<string StateDeviceId, string QueryDeviceId, string Reason>
    {
        (iwSERVICEDeviceKey, iwSERVICEDeviceKey, "IWSERVICE device key for ApplicationState and device-id query"),
        (iwSERVICEDeviceKey, ctx.DeviceId, "IWSERVICE device key for ApplicationState, MDM cert device id for query"),
        (ctx.DeviceId, iwSERVICEDeviceKey, "MDM cert device id for ApplicationState, IWSERVICE device key for query"),
        (ctx.DeviceId, ctx.DeviceId, "MDM cert device id for ApplicationState and query fallback")
    };
}
```

But... The Company Portal needs a "bridge" to
the IME to wake it up!
It needs to start doing its job!





The Install Handoff

Company portal <-> IMEBridge <-> IME

- ManagementExtensionClient
 - Base types
 - Log
 - LogEx
 - ManagementExtensionClient()
 - ManagementExtensionClient(Logger logger, ISemaphoreFactory)
 - AppStatusUpdate(AppInstallStatusReport report) : void
 - CallService<TResult>(Func<IStatusService, Task<TResult>> service)
 - CheckForKeepAlive(object _1, ElapsedEventArgs e) : void
 - CheckInAsync(Guid sessionId) : Task
 - CheckInAvailableAppsAsync(Guid sessionId) : Task
 - ClientChannelClosed(object _1, EventArgs _2) : void
 - ClientChannelFaulted(object _1, EventArgs _2) : void
 - ConnectAsync() : Task<ManagementExtensionConnectionResult>
 - CreateBinding(INamedPipeTransportBindingElementFactory)
 - DecorateLogMessage(string nonLocalizedMessage) : string
 - Disconnect() : void
 - Dispose() : void
 - DownloadProgressUpdate(DownloadProgressReport report) : void
 - GetAllAppsStatusAsync() : Task<AppInstallStatusReport[]>
 - GetAppStatusAsync(Guid applicationId) : Task<AppInstallStatusReport>
 - RestartIdleTracker() : void
 - SetupTimer() : void
 - SyncComplete(SyncResult result) : void

```
private ManagementExtensionClient.LogEx LogException { get; }

private ManagementExtensionClient.Log LogInfo { get; }

public void Dispose()
{
    if (this.disposed)
        return;
    this.Disconnect();
    this.lockConnection.Dispose();
    if (this.ExtendBridgeTimeout)
        this.keepAliveTracker.Stop();
    this.keepAliveTracker.Dispose();
    this.TimeoutExtensionCount = 0;
    this.disposed = true;
}

public async Task<ManagementExtensionConnectionResult> ConnectAsync()
{
    ManagementExtensionClient implementation = this;
    implementation.LogInfo("Initializing connection.");
    implementation.LogInfo($"WCF bridge timeout extension is {implementation.ExtendBridgeTimeout}.");
    IClientChannel clientChannel = (IClientChannel) null;
    if (implementation.ExtendBridgeTimeout)
        implementation.ReceiveTimeout = TimeSpan.Zero;
    if (!await implementation.lockConnection.WaitAsync(ManagementExtensionClient.LockAcquisitionOnConnectTimeout))
    {
        string str = $"Could not acquire a lock on the connection resources within {ManagementExtensionClient.LockAcquisitionOnConnectTimeout}";
        implementation.LogError(str);
    }
}
```

Microsoft.CompanyPortal_11.2.1787.0_x64__8wekyb3d8bbwe

Name	Date modified
IntuneManagementExtensionBridge.exe	4/7/2026 2:46 PM
IntuneManagementExtensionBridge.exe...	6/12/2025 3:18 PM

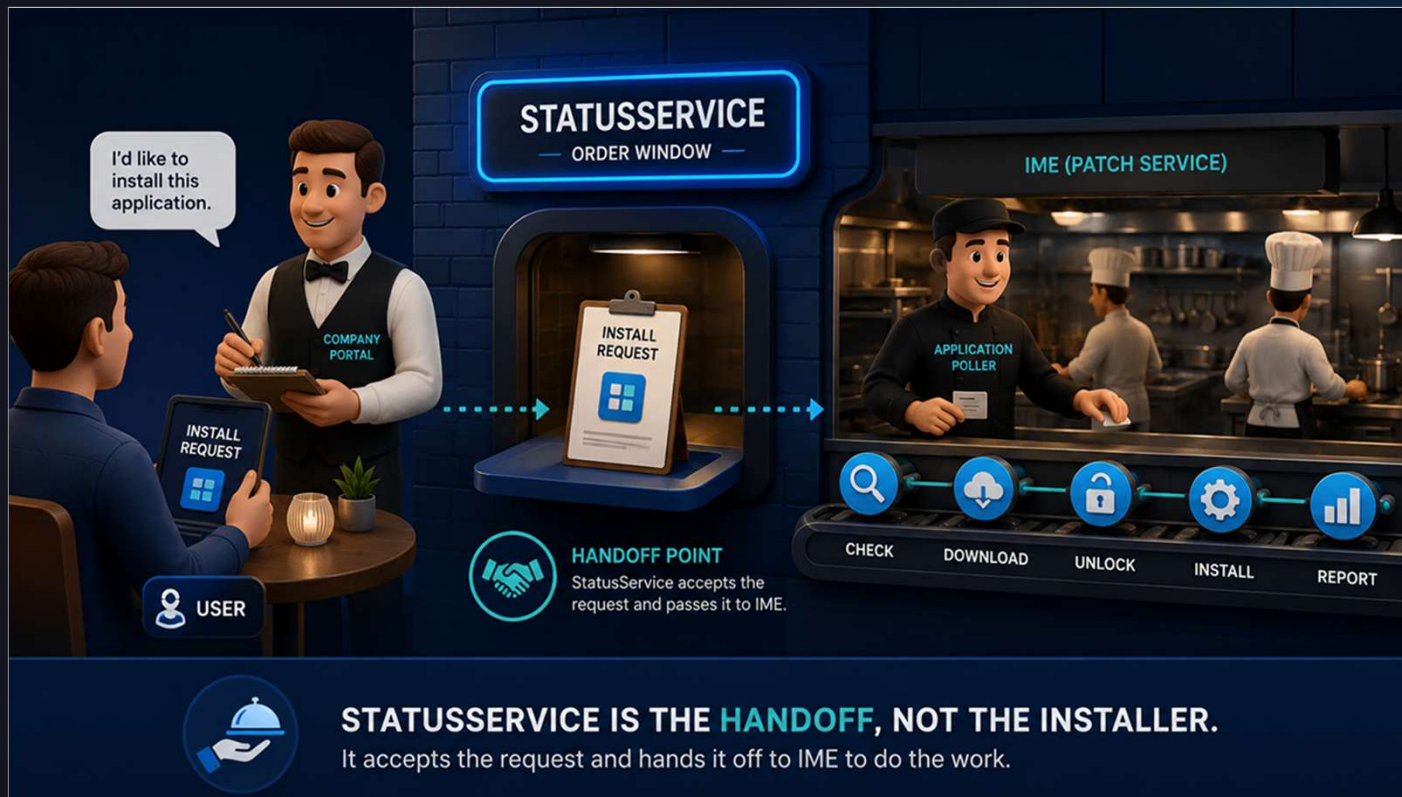
The IMEBridge is the local path Company Portal uses to wake the IME and start the app check in flow.

```
public async Task CheckInAsync(Guid sessionId)
{
    this.LogInfo($"Requesting available and required app {\"M
    int num = await this.CallService<bool>((Func<IStatusServ
    {
        await statusService.CheckInAsync(sessionId);
    }
}
```




The StatusService

The Company Portal performs an App Check-in to kick off the download/install/monitoring



The StatusService nudges the IME and monitors the progress/status

The IME handles the Win32App Install....

```
AllowedImpersonationLevel=Impersonation
Opened StatusService channel: net.pipe://localhost/IntuneManagementExtension/StatusService/
StatusService version=6
```



Using the StatusService Pipe

The Funny thing ... we can do the same thing ourselves..

IME Required App Check-in v1.8

IME Required App Check-in

Mimics the IME part of Company Portal Settings Sync through the IME StatusService named pipe.

Status

IME service: Running

PowerShell: Windows PowerShell 5.1.26100.8115

Elevated: False

StatusService DLL: found

Last result: Accepted, still running or no completion callback within wait window

Actions

Full check-in

Available apps only

Query StatusService state

Fallback service command 200

Fallback compliance command 220

Output

2026-06-08 16:04:44.140 INFO IME Required App Check-in GUI v1.8
2026-06-08 16:04:44.179 INFO Log folder: C:\Users\RudyOoms\AppData\Local\IMERequiredAppCheckinStatusServiceTool\Logs
2026-06-08 16:04:44.183 INFO Use Full check-in to mimic the Company Portal Settings Sync IME app part. Run unelevated, like Company
2026-06-08 16:04:46.610 INFO === AvailableAppsCheckIn ===
2026-06-08 16:04:46.627 INFO IME Required App Check-in StatusService Tool v1.8 started. Action: AvailableAppsCheckIn
2026-06-08 16:04:46.633 DEBUG Log file: C:\Users\RudyOoms\AppData\Local\IMERequiredAppCheckinStatusServiceTool\Logs\IMERequiredAppChe
2026-06-08 16:04:46.637 DEBUG Process is 64-bit: True
2026-06-08 16:04:46.643 WARN Running in 64-bit PowerShell. For this IME StatusService test, 32-bit Windows PowerShell is preferred t
2026-06-08 16:04:46.651 DEBUG Running elevated: False
2026-06-08 16:04:47.273 OK IntuneManagementExtension service status: Running
2026-06-08 16:04:47.291 DEBUG IntuneManagementExtension service path: "C:\Program Files (x86)\Microsoft Intune Management Extension\I
2026-06-08 16:04:47.302 OK Using StatusService library: C:\Program Files (x86)\Microsoft Intune Management Extension\Microsoft.Mar
2026-06-08 16:04:47.346 DEBUG Preparing IME assembly resolver for: C:\Program Files (x86)\Microsoft Intune Management Extension
2026-06-08 16:04:47.358 DEBUG Registered IME assembly resolver.
2026-06-08 16:04:47.369 DEBUG Loaded StatusService assembly: C:\Program Files (x86)\Microsoft Intune Management Extension\Microsoft.I
2026-06-08 16:04:47.373 DEBUG Compiling StatusService client helper against: C:\Program Files (x86)\Microsoft Intune Management Exter
2026-06-08 16:04:47.653 OK StatusService client helper compiled.
2026-06-08 16:04:47.663 INFO Opening IME StatusService named pipe.
2026-06-08 16:04:54.275 INFO SessionId=205dedbc-9668-4c67-804f-a749c21a99c5
2026-06-08 16:04:54.306 INFO ClientIdentity Name=AzureAD\RudyOoms
2026-06-08 16:04:54.314 INFO ClientIdentity User=S-1-12-1-3124292568-1322026969-4294921389-636684985
2026-06-08 16:04:54.327 INFO ClientIdentity Owner=S-1-12-1-3124292568-1322026969-4294921389-636684985
2026-06-08 16:04:54.333 INFO ClientIdentity IsAuthenticated=True, ImpersonationLevel=None
2026-06-08 16:04:54.341 INFO AllowedImpersonationLevel=Impersonation
2026-06-08 16:04:54.348 INFO Opened StatusService channel: net.pipe://localhost/IntuneManagementExtension/StatusService/
2026-06-08 16:04:54.355 INFO StatusService version=6
2026-06-08 16:04:54.360 DEBUG CurrentCheckInId before request=<empty>
2026-06-08 16:04:54.366 INFO Calling AvailableAppCheckInAsync for available apps only

05. The Download + Decryption



First find where SideCar lives

Before the IME can ask the SideCar, it first needs to know where the SideCar lives for this tenant



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- 01 Open Company Portal
- 02 WAM gets access
- 03 Apps appear
- 04 User clicks Install
- 05 IME checks and gets content
- 06 Download, unlock, install, report

First find where SideCar lives

Before asking for content, the client first discovers the right gateway



Discovery happens before ContentInfo. The client must find the right content gateway first.

• The MDM device certificate is used to ask Location Service where SideCar lives

1.2.840.113556.5.11	04 10 9c 5f ad 02 96 36 7b 47 ...
1.2.840.113556.5.14	02 01 02
1.2.840.113556.5.15	02 01 0d
1.2.840.113556.5.16	02 01 2b
1.2.840.113556.5.17	02 01 01
1.2.840.113556.5.18	

04 10 9c 5f ad 02 96 36	...	6
7b 47 8c b3 9b a4 e0 a9	{G...	
ac 9c		

The OID is your tenant id (scrambled around)

```
2026-06-08 14:15:04 Querying discovery endpoint: https://r.manage.microsoft.com/RestUserAuthLocationService/
RestUserAuthLocationService/Certificate/ServiceAddresses
2026-06-08 14:15:05 Found SideCarGatewayService URL: https://agents.amsua0702.manage.microsoft.com/
TrafficGateway/TrafficRoutingService/SideCar/StatelessSideCarGatewayService
```



```

graph TD
    A[IME StatusService] --> B[SideCar (Content Service)]
    B --> C[ContentInfo]
    subgraph ContentInfo
        D[UploadLocation]
        E[DecryptInfo]
    end
  
```




IME Installs and Reports

The Win32App finally gets installed

The installer will be kicked off from the IMECache
It starts installing the Win32App and will report the status

App install
continues

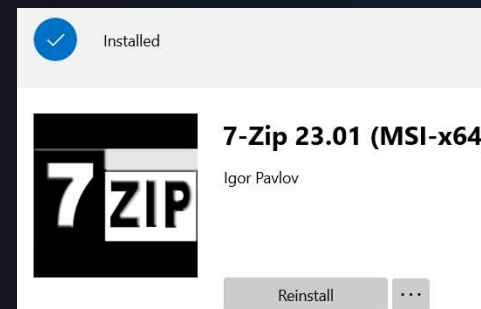
```
[Win32App] Handler invoked with execution type: Install for policy 90eb-66253787efb3 and version: 2.
[Win32App] [GRSManager] Saving GRS values to storage.
[Win32App] [GRSManager] Saved GRS value: 08/29/2023 11:37:36 to path: ddc3cf6e-3b3b-461c-a823-53196902d067\GRS\nmdKzslj7z12yxRyn9wU1hnVY192zxKV...
[Win32App] [Win32AppExecutionExecutor] Using content at: C:\Windows\IMECache\85a3f281-933d-4c72-90eb-66253787efb3_2.
[Win32App] ExecuteWithRetry Parsing InstallEx...
[Win32App] ==Step== Execute retry 0
[Win32App] ==Step== InstallBehavior RegularWin32App, Intent 3, UninstallCommandLine PatchMyPC-ScriptRunner.exe /UninstallPackage
```

Detection
reevaluated

```
[Win32App] Detection script file C:\Program Files (x86)\Microsoft Intune Management Extension\Content\DetectionScripts\85a3f281-933d-4c72-90eb-66253787...
[Win32App] Checked Powershell script result: Detected
[Win32App] Checked Powershell script exitCode: 0 EnforceSignatureCheck: 0 RunAs32Bit: 0 InstallExRu...
[Win32App] Detection script file C:\Program Files (x86)\Microsoft Intune Management Extension\Con...
[Win32App] f applicationDetected: True
```

Toast Success
or Company
Portal update

Which we all
hope to see



Final Summary

Company Portal is a GUI Frontend for the IME



Company Portal

The face for the user.
Shows apps and collects intent.



Works together through
local services and APIs



IME (Intune Management Extension)

The worker on the device.
Executes, installs and reports.

Company Portal

SEARCH MY PC Search apps

Home Apps Downloads & updates Devices Help & support

Apps All Categories

Sort by: Name ascending

Name	Version
7-Zip x64	26.01
Adobe Acrobat Reader x64	26.001.21662
Audacity x64	3.7.8
Camtasia Latest x64	26.1.3.17772
ClickUp x64	3.5.208
DB Browser for SQLite	3.13.1
Dell Command Update for Windows Universal x64	5.7.0

User Action
User clicks Install.
Company Portal creates an install request.

Install Request
(via IWService)

Status & Results
(via StatusService)

Intune Management Extension (IME) Backend Worker



1. App Check-In

IME checks in with Intune to get policy and app assignments.



2. Policy & Content Info

Receives instructions, content location and decrypt information.



3. Download & Decrypt

Downloads the content and uses the device key to decrypt it.



4. Install & Detect

Executes the installer, runs detection rules, and evaluates the result.



5. Report Result

Reports status and details back to Intune via StatusService.